Applied LLMs for Forecasting and Policy Research

Lecture 3: Non-Traditional Data for Nowcasting

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BANK OF CANADA
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Outline

Introduction

Good policy requires timely economic statistics, particularly in crises.

Official statistics often arrive with a lag, so nowcasting attempts to fill the gap.

Problem goes back at least 20 years in academic literature, and there exist well-known public nowcasting models (NY Fed, Atlanta Fed).

One idea is that *non-traditional* data can complement these efforts, most of which rely on traditional data.

National Accounts: Data Quality

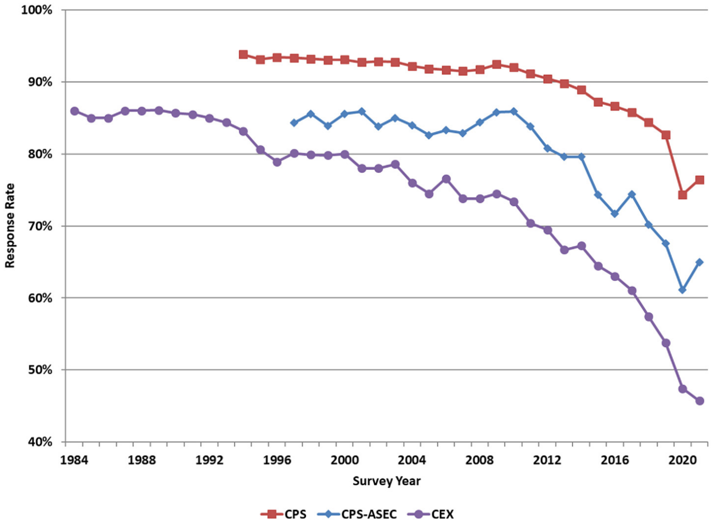
National accounts are vital economic statistics but face pressure:

1. Declining survey participation
2. Budget cuts for national statistics agencies
3. Political interference
4. Pressure for more timely and granular data

In many contexts they are sparse to non-existent [Silungwe et al., 2022]:

1. 1/3 of countries do not publish quarterly accounts (1/2 in Africa)
2. Only four European and five Asian countries produce quarterly accounts within 30 days of the reference period
3. 1/4 of countries have no Household Budget Survey (1/2 in Africa)

[Abraham, 2022]



Naturally Occurring Data

Our main focus here will be on naturally occurring data.

Advantages:

1. Available in real time → tracking impact of shocks on national accounts.
2. Much larger samples than in traditional surveys → ability to obtain more granular measures (e.g. geography, income)
3. Cost of data collection is largely borne by private sector as byproduct of business activity → particularly relevant in lower-income countries.

Main disadvantage is the data collection is "accidental" and has many potential sources of bias and noise.

Outline

Description of concrete application: [Ashwin et al., 2024] with replication materials in
`https://github.com/sekhansen/NowcastingEuroNewsJAE`.

Notebook also illustrates application from Bank of Canada.

Description of random forest algorithm for capturing non-linearities in economic data.

Discussion of [Buda et al., 2022] which uses bank transaction data to directly build national accounts and [Buda et al., 2023] which applies the series to study shock transmission.

Outline

Euro area Growth Nowcasting

[Ashwin et al., 2024] uses daily sentiment indicators from newspapers to nowcast (quarterly) Euro area GDP.

Asks whether sentiment data adds forecasting value beyond benchmark nowcasting models.

Qualified "yes":

1. Particularly early in a quarter before traditional indicators arrive.
2. Affects conditions non-linearly, but non-linear models require sufficient training data.

Newspaper Sources

Table: Total number of articles per country.

	France	Germany	Italy	Spain	All
Sources	Les Échos Le Figaro Le Monde	Die Welt Süddeutsche Zeitung Der Tagesspiegel German Collection	Corriere della Sera La Repubblica Il Sole 24 Ore La Stampa	Expansión El Mundo El País La Vanguardia	
Total articles	1,255,472	833,914	1,497,909	1,407,534	4,994,829

Sentiment Dictionaries

In each quarter, cumulate sentiment through day t using a variety of dictionaries applied to English-language translations of articles.

VADER [Hutto and Gilbert, 2014] detects general-purpose sentiment.

EconLEX [Barbaglia et al., 2025] develops a sentiment dictionary specialized to economics.

Financial stability dictionary from [Correa et al., 2021].

Outline

Supervised Machine Learning Recipe

Begin with observations $(y_i, \mathbf{x}_i)_{i=1}^N$.

Function $f_\theta(\mathbf{x})$ (potentially non-linear) that maps inputs into outputs.

θ is a vector of parameters that determines the shape of the function.

Typical workflow is

1. Split data into training T and test E portions
2. Fit $f_\theta(\mathbf{x})$ on training data to minimize loss function

$$\theta^* = \arg\min_\theta \sum_{i \in T} L(f_\theta(\mathbf{x}_i), y_i)$$

3. Evaluate test-set loss

$$\sum_{i \in E} L(f_{\theta^*}(\mathbf{x}_i), y_i)$$

Tree Models

A tree model is a particular class of non-linear functions. Main idea is to recursively subdivide covariates into subspaces and assign same predicted outcome to all observations in each subspace.

Old method (textbook reference is [Breiman, 1984]) but, until recently, almost no statistical theory.

Individual trees are easy to understand, but somewhat brittle with poor out-of-sample predictive performance.

Ideas to improve trees: bagging, random forests, boosting. All create predictions from multiple trees, sacrifice interpretability for performance.

Definition of a Tree

Let observation i have outcome $y_i \in \mathbf{Y}$ and covariates/features $\mathbf{x}_i \in \mathbf{X}$.

Let $R_1, \ldots, R_M$ be non-overlapping subsets of $\mathbf{X}$ such that $\cup_m R_m = \mathbf{X}$.

A *tree* is a mapping from $\mathbf{X}$ to $\mathbf{Y}$ that takes the form

$$f_\theta(\mathbf{x}) = \sum_{m=1}^{M} \gamma_m \mathbb{1}(x_i \in R_m)$$

where γ_m is a constant.

Parameters of the tree are $\theta = \{R_m, \gamma_m\}_{m=1}^{M}$

$\mathbf{Y} = \mathbb{R}$: *regression* tree

$\mathbf{Y} = \{1, \ldots, K\}$: *classification* tree.

Optimization Problem for Regression Tree

The loss function for a regression tree is

$$L(f_\theta(\mathbf{x}_i), y_i) = (y_i - f_\theta(\mathbf{x}_i))^2$$

Optimization problem is to minimize the RSS with respect to θ.

Conditional on $R_1, \ldots, R_M$, the optimal γ_m is $\overline{y}_m$, the average outcome variable for observations that fall in R_m.

Much harder is to find the optimal division of the covariate space.

Common approach is to use a greedy approximation algorithm that makes successive splits of the data on individual covariates.

Regression Tree Approximation Algorithm

Begin by considering the full covariate space with no splits.

For any variable j, we can define an s and divide observations into those for which $x_{i,j} < s$ (R_1) and $x_{i,j} \geq s$ (R_2).

First step in growing tree is to choose j and s to minimize RSS

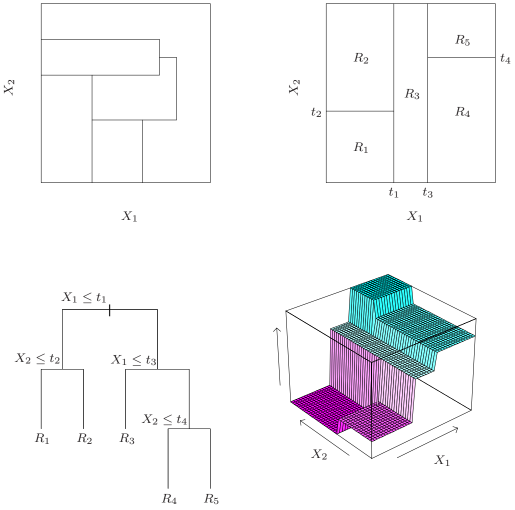
$$\sum_{i:\ \mathbf{x}_i \in R_1} (y_i - \overline{y}_1)^2 + \sum_{i:\ \mathbf{x}_i \in R_2} (y_i - \overline{y}_2)^2$$

Next, repeat the process within each of the two regions and choose a variable and split that most reduce RSS.

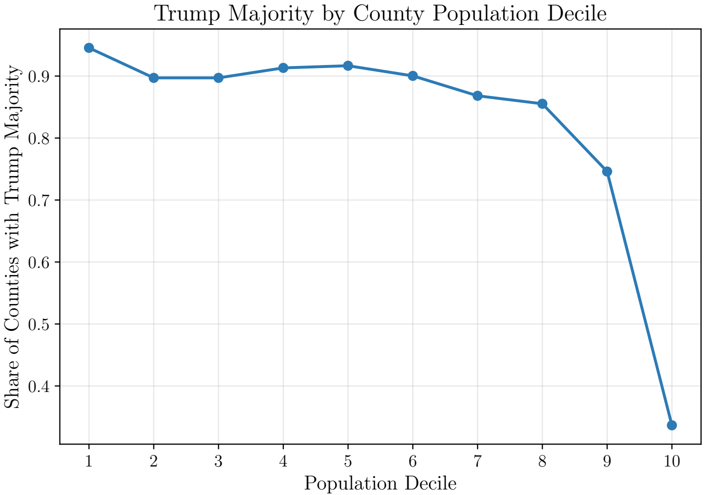
Repeat until stopping criterion is achieved.

The visualization of this splitting process motivates the use of tree terminology. The resulting regions are called *terminal nodes* and the splits are called *internal nodes*.

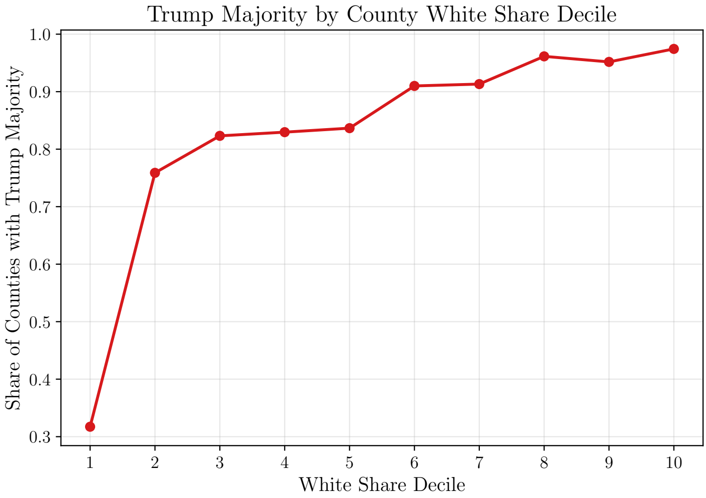
Example from [James et al., 2021]



Trump Win by Population



Trump Win by White Share



Controlling Overfitting

A large tree will explain the data well, but potentially overfit.

In the limit, each observation could in principle be placed in its own node.

To control this, one can use *cost-complexity pruning*.

Idea is to first estimate a large tree f_0 and then prune it back to a tree $f \subset f_0$ that maximizes

$$\sum_{m=1}^{M(f)} \sum_{i:\mathbf{x}_i \in R_m} (y_i - \overline{y}_m)^2 + \alpha M(f)$$

where $M(f)$ is number of terminal notes in f and $\alpha > 0$ controls penalization strength.

Why start with large tree and prune back? Early splits may not reduce RSS in themselves, but lead to further informative splits.

Selecting Penalty Term

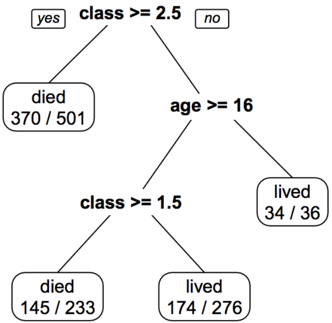
Each value of α is associated with some subtree $f_\alpha \subset f_0$, so one can generate a sequence of subtrees for different values of the penalty term.

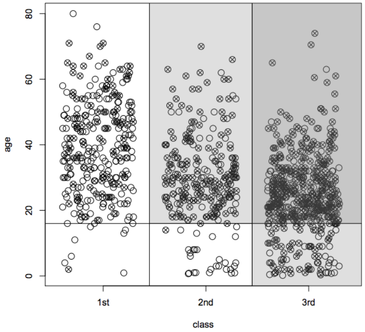
Use k-fold cross validation and, for each fold, estimate a sequence of trees on the training data as a function of α and compute RSS in the validation data for each tree.

Compute value α^* that minimizes average loss in held-out data.

Select f_{α^*} from the original sequence.

Example from [Varian, 2014]





Combining

Regression/classification trees tend to have high variance: estimating on a new draw of training data is likely to produce different splits.

This makes the out-of-sample predictive performance of individual trees worse than that of other well-known machine learning methods.

One solution is to combine the predictions of many trees to average out the noise in any particular tree.

Most popular ways of combining trees:

1. Bagging
2. Random Forests
3. Boosting

Example of *ensemble* methods.

Bagging

Bagging, or bootstrap aggregation, repeatedly resamples training data and estimates a separate regression tree for each draw.

Let $\hat{f}_s(\mathbf{x}_i)$ be the predicted outcome for y_i associated with the tree estimated from the sth bootstrap sample.

The overall prediction for y_i is the simple average

$$\frac{1}{S} \sum_s \hat{T}_s(\mathbf{x}_i)$$

Individual trees are not pruned, instead averaging reduces their noise.

Bagging can be used for any high-variance method to reduce noise.

Random Forest

The random forest algorithm is an extension of bagging.

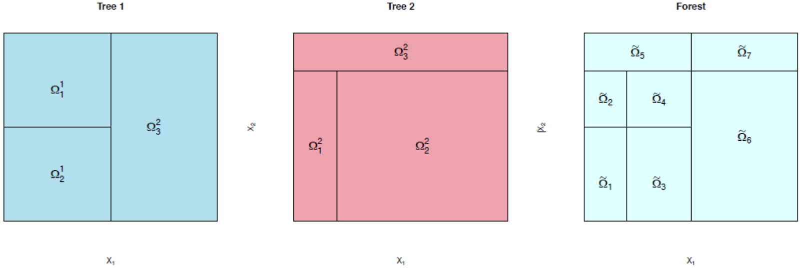
At each split, we only consider a randomly drawn subset of the J available variables; typical number of variables is $\sqrt{J}$.

Has the effect of de-correlating predictions across bootstrap draws in the presence of strong predictors.

Averaging less correlated predictions is more effective at reducing prediction error than plain bagging.

Why Forests?

Besides potential variance reduction, combining trees approximates smooth functions better.



Outline

Background

[Buda et al., 2022] applies national accounting principles (from ESA 2010) to organize large-scale spending data from major private sector bank (BBVA).

Results in a consumption panel that:

1. Aggregates to national accounts consumption at comparable frequencies
2. Has almost two million participants
3. Daily frequency since 2015, updated in real time
4. Breakdown of consumption into COICOP categories

Applications: distributional national accounts for consumption, rich characterization of consumption growth, high-frequency response to monetary policy shocks [Buda et al., 2023].

Data

All BBVA-mediated transactions from 2015Q1 through present.

Panel of 1.8 million "active" customers who regularly use accounts and are not self-employed.

Three billion individual transactions with total value of 200 billion euros.

Compared to Spanish Household Budget Survey (HBS):

1. Observed rather than reported spending
2. (Much) larger data both in cross section and time series
3. Biased sample with respect to Spanish population
4. Possibility of non-BBVA-mediated consumption

Households

We consider BBVA clients linked in a household whenever:

1. They have co-signed a financial contract (bank account, loan, mortgage, etc.)
2. They reside in the same postal code

Households include both active and non-active clients.

Consumption vs. Non-Consumption Spending

Not all financial outflows are consumption.

We first limit attention to outflows to firms/organizations with tax ID.

We use ESA 2010 principles to design appropriate filters, for example:

1. House repairs
2. Payments to financial institutions
3. Tax payments
4. Real estate purchases

Structure of Payments Data

Filters are implemented based on transaction metadata:

1. Card Spending: Merchant Client Codes
2. Direct Debits: Internal BBVA codes
3. Irregular Transfers: Counterparty Firm Tax ID + NACE

Metadata also used for allocating COICOP categories.

Category	Description
01	Food and Non-Alcoholic Beverages
02	Alcoholic Beverages, Tobacco, and Narcotics
03	Clothing and Footwear
04	Housing, Water, Electricity, Gas, and Other Fuels
05	Furnishings, Household Equipment, and Routine Household Maintenance
06	Health
07	Transport
08	Communication
09	Recreation and Culture
10	Education
11	Restaurants and Hotels
12	Miscellaneous Goods and Services

Table 1: COICOP Consumption Categories (Two-Digit)

Spending Category	Volume of Transactions	Number of Transactions
Offline Card Transactions	60,319 million €	1,772 million
Online Card Transactions	11,858 million €	313 million
Direct Debits	66,036 million €	752 million
Cash Withdrawal	64,592 million €	359 million
Transfers excl. rent	11,148 million €	15 million

Housing Consumption

Housing is major component of consumption, but is largely imputed.

Build household rental payments using search of text description.

In line with ESA guidelines, regress on geographic location, joint income, joint utility spending.

Good fit in-sample and out-of-sample. Impute rental payments to whole sample.

Filtering is Quantitatively Important

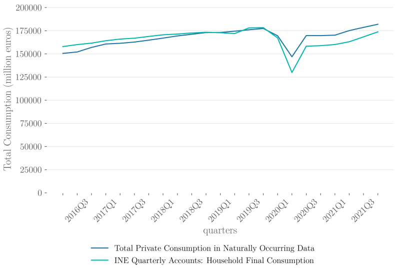
	Volume of Transactions (2019)
(1) Cash Withdrawal	10,142 million €
Outflows under concepts of Card / Transfer / Direct Debit	51,342 million €
Out of which: Outflows to organizations	31,232 million €
(2) Out of which: Consumption-related transactions	23,959 million €
(3) Imputed Rent	8,158 million €
Total unweighted consumption (1) + (2) + (3)	42,259 million €

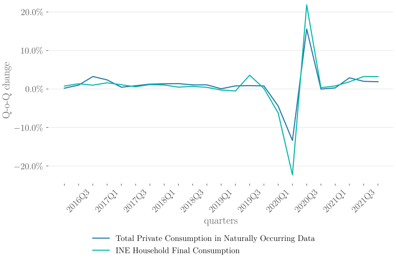
Table 5: Impact of filtering spending to consumption (2019)

Weighting

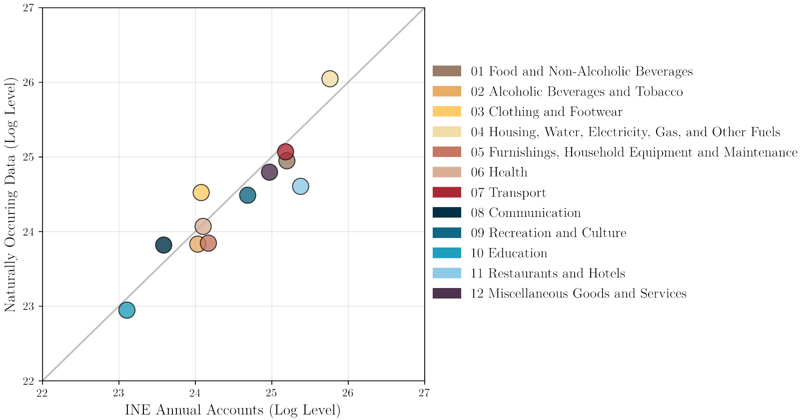
$$c_i = \frac{\sum_{j \in A(i)} c_j^{\text{NH}} + c_{h(i)}^{\text{H}}}{A(i) + 0.5O(i)} \quad c_{t;g,a,q}^{W} \equiv c_{t;g,a,q} \times \left(\frac{x_{g,a,q}^{\text{INE}}}{x_{\tau(t);g,a,q}^{\text{BBVA}}} \right)$$

Aggregate Quarterly Consumption

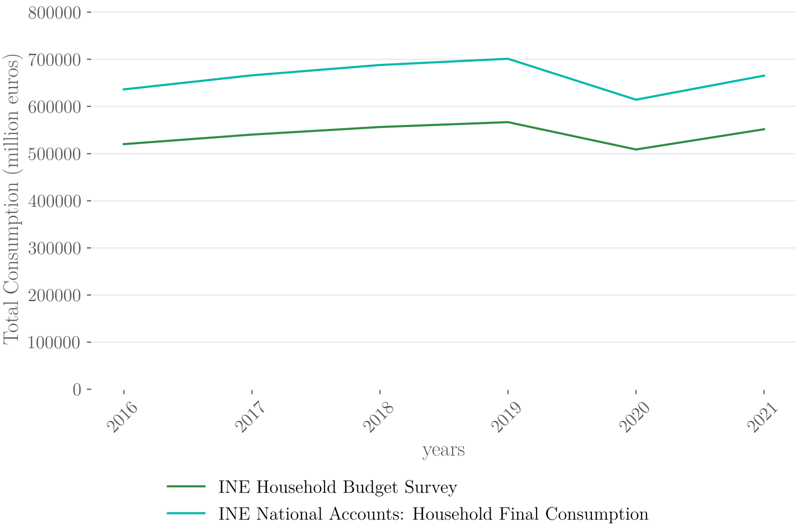




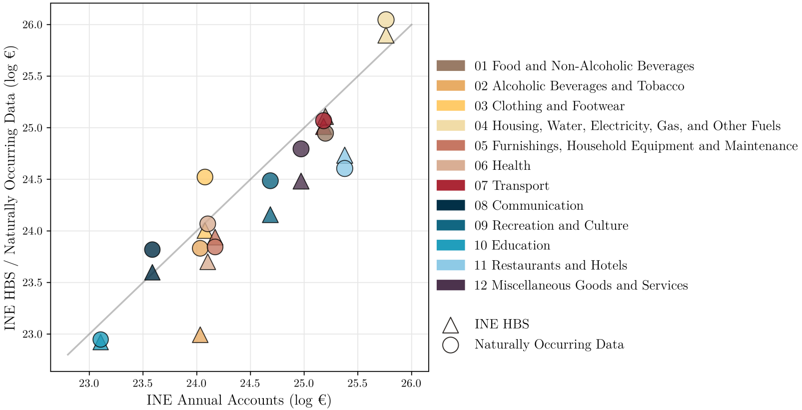
Coverage by Consumption Category (2019)



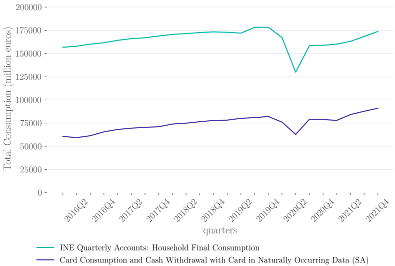
Comparison to Household Budget Survey

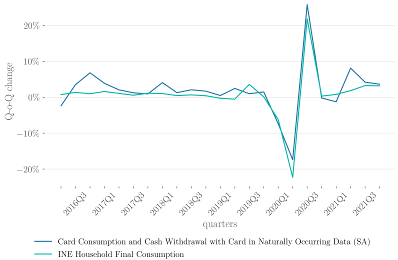


Coverage by Consumption Category (HBS)

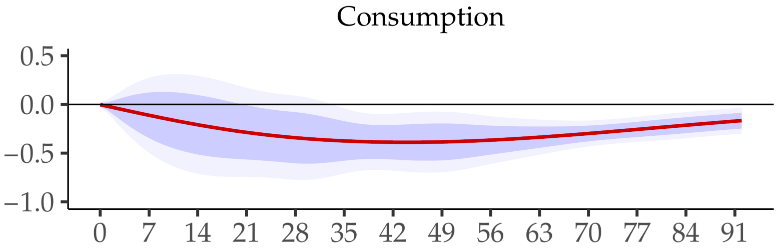


Comparison to Card Data





Consumption Response to Monetary Policy Shock [Buda et al., 2023]



References I

Abraham, K. G. (2022).
Big Data and Official Statistics.
Review of Income and Wealth, forthcoming.

Ashwin, J., Kalamara, E., and Saiz, L. (2024).
Nowcasting Euro area GDP with news sentiment: A tale of two crises.
Journal of Applied Econometrics, 39(5):887–905.

Barbaglia, L., Consoli, S., Manzan, S., Pezzoli, L., and Tosetti, E. (2025).
Sentiment analysis of economic text: A lexicon-based approach.
Economic Inquiry, 63:125–143.

Breiman, L. (1984).
Classification and Regression Trees.
Routledge, Boca Raton, Fla., 1st edition edition.

Buda, G., Carvalho, V. M., Corsetti, G., Duarte, J. B., Hansen, S., Ortiz, Á., Rodrigo, T., and Rodríguez Mora, J. V. (2023).
Short and Variable Lags.
SSRN Electronic Journal.

References II

Buda, G., Carvalho, V. M., Hansen, S., Mora, J. V. R., Ortiz, A., and Rodrigo, T. (2022).
National Accounts in a World of Naturally Occurring Data: A Proof of Concept for Consumption.
Technical Report 2244, Faculty of Economics, University of Cambridge.

Correa, R., Garud, K., Londono, J. M., and Mislang, N. (2021).
Sentiment in Central Banks' Financial Stability Reports.
Review of Finance, 25(1):85–120.

Hutto, C. and Gilbert, E. (2014).
VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text.
Proceedings of the International AAAI Conference on Web and Social Media, 8(1):216–225.

James, G., Witten, D., Hastie, T., and Tibshirani, R. (2021).
An Introduction to Statistical Learning: With Applications in R.
Springer, New York NY, 2nd ed. 2021 edition edition.

References III

Silungwe, A., Baer, A., and Guerreiro, V. (2022).
2020 Global Stocktaking of National Accounts Statistics: Availability for Policy and Surveillance.
IMF Working Papers.

Varian, H. R. (2014).
Big Data: New Tricks for Econometrics.
Journal of Economic Perspectives, 28(2):3–28.